Logic Realism Theory

Non-Decomposability, Entanglement, and Bell's Theorem

Derived from $A\Omega = L_3(I\infty)$

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Abstract

This document consolidates the derivation of quantum non-decomposability and Bell's theorem from Logic Realism Theory's single foundational equation $A\Omega = L_3(I\infty)$. LRT is a one-equation theory. Every result here is downstream of that equation. No independent primitives are introduced. The argument proceeds in four stages. First, the Unique Necessary Successor (UNS) constraint is derived as a theorem of L_3 — not posited independently — because a complete L_3-constrained state admits exactly one admissible successor by Identity and Excluded Middle alone. Second, quantum probability is reinterpreted as epistemic underdetermination of state description, with the Born rule reinterpreted via Gleason as the unique consistent measure over incomplete descriptions. Third, the Joint Resolution Condition (JRC) is defined as the L_3-criterion for configuration irreducibility, and the JRC-Schmidt Lemma proves JRC is equivalent to Schmidt rank > 1. Fourth, the Non-Decomposability Theorem shows that decomposing a JRC-satisfying configuration into local sub-descriptions necessarily violates Excluded Middle. Bell's theorem follows as downstream consequence. Two open formalizations are identified for Lean 4 development.

1. Foundational Architecture

1.1 The Core Equation

All formalism derives from:

$$A\Omega = L_3(I\infty)$$

$I\infty$ is the Infinite Information Space: the totality of all logically coherent distinguishable configurations. L_3 is the Three Fundamental Laws of Logic (Identity, Non-Contradiction, Excluded Middle) acting as a selection functional. $A\Omega$ is the currently instantiated, L_3-admissible subset. Actualization is inclusion: $A\Omega \subseteq I\infty$.

1.2 Three Levels of Structure

The GPT review prompted a refinement that sharpens the ontological architecture. Three levels are now distinguished:

$$I\infty \qquad \text{total configuration domain}$$

$$\mathscr{A} = L_3(I\infty) \qquad \text{admissible domain (all } L_3\text{-satisfying configurations)}$$

$$\Gamma_A\Omega \subset \mathscr{A} \qquad \text{single actualized trajectory}$$

$\mathscr{A}$ is the admissible domain — what L_3 selects from I∞. Γ_AΩ is the actualized sequence — the single trajectory of necessary successors through $\mathscr{A}$. These are distinct. The core equation defines $\mathscr{A}$. The trajectory is what physics traces.

1.3 The Transition Operator and UNS as Theorem

Time is logical succession. The transition operator T governs permissible successors over $\mathscr{A}$:

$$T : \mathscr{A} \times \mathscr{A} \to \{0,1\}$$

$$T(s_n, s_\{n+1\}) = 1 \quad \text{iff} \quad s_\{n+1\} \text{ is the logically necessary successor to } s_n$$

The Unique Necessary Successor (UNS) constraint is not an independent postulate. It is a theorem of L_3:

THEOREM (UNS)	∀ s_n ∈ $\mathscr{A}$, ∃! s_{n+1} ∈ $\mathscr{A}$ such that T(s_n, s_{n+1}) = 1

Proof sketch: A complete state description resolves every relevant predicate under Excluded Middle. L_3 acting on a fully resolved state admits exactly one configuration satisfying all three constraints simultaneously as successor.

Multiple successors would require one of two things, both inadmissible: (1) a predicate unresolved in s_n — meaning the description is incomplete, not that multiple successors are genuinely open; or (2) two distinct configurations both fully satisfying L_3 under identical constraints — which violates Identity, because two configurations satisfying all the same constraints under the same respect are indistinguishable and therefore identical, not two.

Therefore: apparent multiplicity is always an artifact of incomplete state description. At the level of fully L_3-specified states, exactly one successor is logically necessary. UNS is derived, not assumed.

ESTABLISHED	UNS follows from L_3 alone. It is not a third primitive alongside L_3 and I∞. LRT is a one-equation theory: AΩ = L_3(I∞). UNS, temporal ordering, measurement definiteness, and entanglement correlations are all downstream of that single equation.

2. Quantum Probability as Epistemic Underdetermination

2.1 The Standard Problem

If T produces unique necessary successors, what is quantum probability? The apparent randomness of quantum measurement is not in dispute empirically. The question is its interpretation.

LRT's answer: the transition amplitude T(s_n, s_{n+1}) is not a measure over ontologically open alternatives. It is an epistemic measure — it quantifies how much of the full L_3-constrained interaction structure is captured by our state description.

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T(s_n, s_{n+1}) = 1   iff   s_n is fully L₃-constrained (complete
                            description)

T(s_n, s_{n+1}) < 1   iff   s_n is incompletely specified
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2.2 Completeness Criterion

A state description s is complete with respect to an interaction iff every predicate relevant to that interaction is determinately resolved under Excluded Middle:

> *s is complete iff: ∀ predicate P relevant to the successor interaction, P ∈ s is determinately resolved.*

Incompleteness is not a feature of the world. It is a feature of the description relative to the interaction scope. The world at every stage has a complete state. Our descriptions are always projections — subsets of full predicate resolution.

2.3 Gleason Reinterpreted

Gleason's theorem states: the only consistent probability measure on the closed subspaces of a Hilbert space of dimension ≥ 3 is the Born rule measure Tr(ρP).

Standard reading: this gives the unique way to assign probabilities to quantum outcomes.

LRT reinterpretation: Gleason gives the unique consistent measure over incompletely specified states in $\mathscr{A}$. It is not a measure over ontological possibilities. It is the unique coherent way to represent epistemic underdetermination of state description within the Hilbert space structure that L_3 already forced via the Masanes-Müller derivation.

The Born rule is therefore doubly grounded in LRT:

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Derivation:      L₃ → H (Masanes-Müller) → Born rule (Gleason)

Interpretation:  Born rule = unique consistent measure over
                             specification deficit
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These are not competing accounts. The derivation gives the form; the reinterpretation gives the referent.

ESTABLISHED	Gleason's theorem is preserved in full. What changes is its referent: it describes our epistemic situation with respect to incomplete descriptions, not the ontological branching structure of the world. This reinterpretation is consistent with the Masanes-Müller derivation chain already in LRT.

3. The Joint Resolution Condition

3.1 Definition

The Joint Resolution Condition (JRC) is the L_3-criterion for configuration irreducibility. It identifies when a composite configuration cannot be decomposed into independent local sub-descriptions without violating L_3.

DEFINITION	JRC holds for s_AB $\in \mathscr{A}$ iff $\exists$ predicate P such that: (1) P is well-formed under L_3; (2) P is not resolvable from s_A alone; (3) P is not resolvable from s_B alone; (4) P is resolvable from s_AB.

When JRC holds, s_AB is L_3-irreducible. Attempting to decompose it into s_A and s_B independently leaves a well-formed predicate without determinate resolution — violating Excluded Middle in both sub-descriptions simultaneously.

3.2 When JRC Holds and When It Does Not

Not all composite configurations satisfy JRC. A configuration of two non-interacting classical particles has no relational predicates whose resolution requires the joint description. Each particle's predicates resolve independently.

JRC is satisfied when the interaction structure of the composite system generates predicates that are irreducibly relational — predicates whose truth conditions reference both subsystems jointly and cannot be factored into independent local conditions.

In quantum mechanics, these are precisely the predicates corresponding to entanglement witnesses — operators whose expectation values on entangled states cannot be reproduced by any product state.

3.3 Excluded Middle Violation Under Decomposition

If JRC holds and we attempt to decompose s_AB into independently L_3-complete sub-descriptions s_A and s_B:

L_3-completeness of s_A requires: every well-formed predicate on A is determinately resolved.

L_3-completeness of s_B requires: every well-formed predicate on B is determinately resolved.

But the JRC predicate P is well-formed under L_3 and not resolvable from either. Therefore P lacks a determinate truth value in both s_A and s_B. Excluded Middle is violated in both sub-descriptions simultaneously.

This is not merely a practical limitation. It is a structural L_3 failure.

4. JRC-Schmidt Lemma

4.1 Setup

Let H_AB = H_A ⊗ H_B with dim(H_A) = m ≥ 2 and dim(H_B) = n ≥ 2. Every vector ψ_AB in H_AB has a Schmidt decomposition:

$$|\psi_AB\rangle = \Sigma_i\ \lambda_i\ |a_i\rangle \otimes |b_i\rangle, \quad \lambda_i > 0, \quad \Sigma_i\ \lambda_i^2 = 1$$

The Schmidt rank r(ψ) is the number of nonzero λ_i. A state is entangled iff r(ψ) > 1.

4.2 The Lemma

LEMMA	JRC-Schmidt Correspondence: s_AB ∈ $\mathscr{A}$ with Hilbert space representative |ψ_AB⟩ satisfies JRC if and only if r(ψ) > 1.

4.3 Proof — Forward Direction (JRC implies Schmidt rank > 1)

Suppose JRC holds for s_AB. Then there exists a predicate P well-formed under L_3 whose resolution requires the joint configuration.

In Hilbert space representation, P corresponds to a Hermitian operator M_AB on H_AB that is not of the form M_A ⊗ I or I ⊗ M_B.

Suppose for contradiction that r(ψ) = 1. Then:

$$|\psi_AB\rangle = |a\rangle \otimes |b\rangle$$

For a product state, define the reduced operators:

$$M_A^{eæa} = Tr_B[(I \otimes |b\rangle\langle b|)\ M_AB]$$

For any operator M_AB, the expectation value on a product state factorizes through local reduced operators. Therefore every predicate on s_AB is resolvable from the local descriptions alone.

This contradicts the assumption that JRC holds. Therefore r(ψ) > 1. ∎

4.4 Proof — Backward Direction (Schmidt rank > 1 implies JRC)

Suppose r(ψ) > 1. We exhibit a concrete JRC-satisfying predicate.

Since r(ψ) > 1, write:

$$|\psi_{AB}\rangle = \lambda_1|a_1\rangle\otimes|b_1\rangle + \lambda_2|a_2\rangle\otimes|b_2\rangle + \ldots$$

Define the correlation operator:

$$C_{AB} = |a_1\rangle\langle a_2| \otimes |b_1\rangle\langle b_2| + |a_2\rangle\langle a_1| \otimes |b_2\rangle\langle b_1|$$

More generally, take the CHSH operator on the two-qubit subspace spanned by {|a_1⟩, |a_2⟩} ⊗ {|b_1⟩, |b_2⟩}. The CHSH operator has expectation value exceeding 2 on any entangled state and at most 2 on any product state — this is the mathematical content of Bell's theorem.

Therefore C_AB is: (1) well-formed under L_3 (it is Hermitian with determinate eigenvalues); (2) not resolvable from s_A or s_B alone (its expectation value exceeds the local bound); (3) resolvable from s_AB. JRC holds. ∎

ESTABLISHED	JRC ⇔ Schmidt rank > 1. Both directions proved. The correspondence is exact: the class of JRC-satisfying configurations in $\mathscr{A}$ is identical to the class of entangled states in H_AB.

5. Entangled States Are Forced

5.1 The Forcing Argument

The JRC-Schmidt Lemma establishes correspondence. But we need more: JRC-satisfying states must be shown to necessarily exist, not merely to be possible.

5.2 Theorem: Schmidt Rank > 1 States Are Forced

THEOREM	For any composite system with H_AB = H_A ⊗ H_B and dim(H_A) ≥ 2, dim(H_B) ≥ 2, the space H_AB necessarily contains vectors with Schmidt rank > 1.

Proof: This is a theorem of linear algebra. The product states — those with Schmidt rank 1 — are the set of decomposable tensors in H_A ⊗ H_B. This set has dimension m + n − 1 (as a variety), while H_AB has dimension m × n. For m, n ≥ 2, m × n > m + n − 1. Therefore the product states are a measure-zero proper subset of H_AB. Their span is the full H_AB, but they do not exhaust it. H_AB necessarily contains non-product vectors. ∎

The premises — tensor product structure and dim ≥ 2 for each subsystem — are both derived in LRT:

Tensor product structure: from local tomography [Theorem 6.2, existing], which is derived from Determinate Identity [Theorem 1, existing].

dim ≥ 2: from Bit Equivalence [MM4], which is derived from Determinate Identity and continuous reversibility in the existing Hilbert space paper.

Therefore: entangled states exist in H_AB as a matter of logical necessity, not empirical contingency. Their existence is forced by the same derivation chain that gives complex Hilbert space from L_3.

ESTABLISHED	Entangled states are not merely permitted by L_3 (as Theorem 3 in the existing Hilbert space paper shows). They are forced by the tensor product structure that local tomography requires. This upgrades the existing permissibility result to a necessity result.

6. The Non-Decomposability Theorem

6.1 Statement

THEOREM	Non-Decomposability under L_3: Let s_AB $\in \mathscr{A}$ be a configuration with Schmidt rank r(ψ) > 1. Then there is no L_3-complete decomposition s_AB = f(s_A, s_B) for any function f that preserves L_3-completeness of the sub-descriptions s_A and s_B.

6.2 Proof

Suppose for contradiction that s_AB = f(s_A, s_B) with both s_A and s_B L_3-complete.

L_3-completeness of s_A: every well-formed predicate on A is determinately resolved under Excluded Middle.

L_3-completeness of s_B: every well-formed predicate on B is determinately resolved under Excluded Middle.

By the JRC-Schmidt Lemma (backward direction): since r(ψ) > 1, there exists a predicate C_AB well-formed under L_3 whose resolution requires the joint configuration — it is not resolvable from s_A or s_B alone.

Now consider what the decomposition f(s_A, s_B) implies. If s_AB is fully determined by f acting on s_A and s_B, then every predicate on s_AB must be derivable from f(s_A, s_B).

Case 1: f is local. Then f(s_A, s_B) can only encode information already present in s_A and s_B independently. Since C_AB is not resolvable from either alone, it is not resolvable from any local f.

Case 2: f is nonlocal. Then f encodes additional correlation information beyond what is present in s_A and s_B individually. But if f generates information not present in its inputs, then s_A and s_B are not a complete description of s_AB — f is supplying the missing content. This means s_A and s_B are not a complete basis for s_AB, contradicting the assumption that they are L_3-complete sub-descriptions of s_AB.

In both cases: C_AB is a well-formed predicate under L_3 that lacks determinate resolution in the decomposed description. Excluded Middle is violated.

Contradiction. No L_3-complete decomposition of s_AB into local sub-descriptions exists. ∎

6.3 What the Proof Establishes

The proof shows that any attempt to restore local independent descriptions for an entangled configuration necessarily violates Excluded Middle. This is not a contingent empirical fact about quantum systems. It is a structural consequence of L_3 applied to JRC-satisfying configurations.

Bell's hidden variable program attempts exactly the decomposition the theorem rules out: it tries to complete local descriptions s_A and s_B with parameters λ that restore local determinism. The Non-Decomposability Theorem shows this is not merely empirically inadequate but logically inadmissible.

ESTABLISHED	Conditional on Open 1 and Open 2 below. The philosophical argument is sound. The two open formalizations are required before this reaches Lean 4 proof status.

7. Bell's Theorem as Downstream Consequence

7.1 What Bell Proves

Bell's theorem: no theory satisfying local realism can reproduce all predictions of quantum mechanics. Local realism requires that measurement outcomes on separated systems are determined by local variables alone, with no superluminal influence.

Formally: Bell assumes that for any entangled pair, there exist complete local state descriptions λ_A and λ_B such that joint statistics factor as:

$$P(a,b|\lambda) = P_A(a|\lambda_A) \times P_B(b|\lambda_B)$$

Bell then shows that any such theory satisfies the CHSH inequality:

$$S = |E(a,b) - E(a,b') + E(a',b) + E(a',b')| \leq 2$$

Quantum mechanics violates this bound, achieving $S = 2\sqrt{2}$ for maximally entangled states.

7.2 Where Bell's Assumption Fails Under LRT

Bell requires that each wing of an entangled experiment has a complete local description. LRT's Non-Decomposability Theorem establishes that for JRC-satisfying configurations, no such complete local description exists.

The comparison is precise:

Framework	What the entangled system requires
Bell (hidden variables)	s_A ⊗ s_B + λ_A + λ_B → determinate joint outcome
LRT	s_AB (L_3-irreducible joint config) → determinate joint outcome. s_A and s_B are incomplete projections of s_AB.

Bell's theorem rules out theories of the first kind. LRT is a theory of the second kind. The locality assumption Bell's proof requires — that each subsystem has a complete local description — is precisely what the Non-Decomposability Theorem denies for entangled configurations.

LRT does not violate Bell's theorem. It sidesteps the assumption the proof requires.

7.3 The Stronger Claim

Standard responses to Bell either: (a) accept nonlocality as a brute fact (Copenhagen, many-worlds), (b) embrace explicit nonlocality (Bohmian mechanics), or (c) deny hidden variables while accepting indeterminism (orthodox QM).

LRT offers a fourth option: the demand for complete local descriptions is not merely empirically unachievable. It is logically inadmissible. For JRC-satisfying configurations, Excluded Middle itself forbids the decomposition Bell requires.

Bell's theorem is then not the deepest result in this area. It is a downstream consequence of a more fundamental constraint: the non-decomposability of L_3-irreducible configurations in $\mathscr{A}$.

ESTABLISHED	Bell's locality assumption is L_3-inadmissible for entangled systems. This follows from Non-Decomposability, which follows from JRC-Schmidt, which follows from tensor product structure, which follows from local tomography, which follows from Determinate Identity. Bell is downstream of L_3.

8. Comparison with Deterministic Hidden-Variable Programs

8.1 The Standard Comparison

LRT is sometimes placed alongside Bohmian mechanics and 't Hooft's cellular automaton model as a deterministic program. The comparison is instructive but the distinction is substantive. Conflating them misrepresents what LRT is doing foundationally.

8.2 What Bohmian Mechanics Does

Bohmian mechanics adds a pilot wave to the standard quantum formalism. Particles have definite positions at all times, guided by a wave function satisfying the Schrödinger equation. The grounding direction is: quantum formalism first, then supplement with

hidden structure to restore determinism. Bell's theorem is accommodated by accepting nonlocality as a brute fact.

8.3 What 't Hooft's Program Does

't Hooft's cellular automaton interpretation postulates a deterministic substrate below the quantum level. Quantum states are equivalence classes of underlying deterministic states. The grounding direction is again: quantum formalism primary, hidden substrate added underneath it. The substrate is posited, not derived.

8.4 What LRT Does

LRT does neither. It derives quantum structure from logical constraints on what can be instantiated. The grounding direction is opposite: L_3 first, physics downstream. No pilot wave is added. No hidden substrate is posited. The Hilbert space structure of quantum mechanics is derived via Masanes-Müller. The Born rule is derived via Gleason. Entanglement and Bell violations are derived via JRC and Non-Decomposability.

Program	Grounding Direction	Hidden Structure?
Bohmian mechanics	QM → hidden variables	Yes — pilot wave added
't Hooft (CA)	QM → hidden substrate	Yes — substrate posited
LRT	L_3 → QM	No — QM structure derived

8.5 The Constraint vs. Variable Distinction

A variable has a value. Hidden variable theories posit pre-existing definite values that determine outcomes. A constraint is a rule. L_3 does not assign values to configurations. It specifies which configurations are admissible. Measurement reveals which admissible configuration is actualized. No hidden information is consulted. The constraint was constitutive of the state throughout.

Bell rules out local hidden variables — pre-existing definite values that determine outcomes. LRT does not posit such values. Non-decomposability of entangled states is a structural feature of L_3-admissible composite configurations, not a consequence of hidden value assignments. Bell's theorem does not threaten LRT because LRT is not a hidden variable theory.

ESTABLISHED	LRT is not a hidden variable theory, not a pilot wave theory, not a substrate theory. It derives quantum structure from logical constraints rather than supplementing quantum structure with additional ontology. The grounding direction — from logic to physics — distinguishes it from all programs in this vicinity.

9. The UNS Proof and Leibniz Identity of Indiscernibles

9.1 The Gap in the Proof Sketch

The UNS proof sketch in Section 1.3 argues that multiple successors would require either an unresolved predicate (incomplete description) or two distinct configurations both satisfying L_3 under identical constraints. A reviewer will press: why cannot two numerically distinct configurations satisfy all the same L_3 constraints?

9.2 Leibniz Identity of Indiscernibles

Leibniz's principle: entities sharing all the same properties are identical — not merely similar but numerically one. L_3's Identity law grounds this ontologically. A configuration is determinately what it is. Two configurations satisfying all the same L_3 constraints under the same respect are not two configurations. They are one configuration described twice.

Formally: let s and s' both be L_3-admissible successors to s_n under constraint set C(s_n). If s and s' are numerically distinct, Identity requires they differ in at least one predicate. But if they satisfy all the same constraints under C(s_n), no such predicate exists. Therefore s = s'. The appearance of two successors is always either incomplete specification of s_n or mislabeling of one configuration under two descriptions.

9.3 Tightened UNS Theorem

THEOREM (UNS — Tightened)	∀ s_n ∈ $\mathscr{A}$ with complete L_3-constrained description, ∃! s_{n+1} ∈ $\mathscr{A}$ such that T(s_n, s_{n+1}) = 1. Existence: L_3 acting on a fully resolved state admits at least one admissible successor; otherwise the admissible domain collapses to empty, contradicting the foundational equation. Uniqueness: any two admissible successors satisfying all the same constraints in C(s_n) are identical by Leibniz Identity via L_3's Identity law. One successor, not two. ∎

Status: ESTABLISHED as proof sketch. Lean 4 formalization is Open 3 below.

10. Quantum Contextuality as Downstream Consequence

10.1 Kochen-Specker

The Kochen-Specker theorem (1967) proves that in a Hilbert space of dimension ≥ 3, there is no consistent assignment of definite values to all quantum observables simultaneously that is context-independent — independent of which other compatible observables are co-measured. This is stronger than Bell: it applies to any single quantum system of sufficient dimension, not just entangled pairs.

A derivation of Bell from ${L3} that does not address Kochen-Specker is incomplete. The following shows contextuality is also downstream of ${L3} via UNS and JRC.

10.2 Context as Interaction Scope

Under LRT, a measurement context is an interaction scope: the set of predicates relevant to a given interaction. From Section 2.2, a state description is complete relative to an interaction scope. Different measurement contexts define different interaction scopes. The same configuration s_n can have different predicate resolutions under different contexts without contradiction, because predicates are context-indexed. Predicate P in context C_1 and predicate P in context C_2 are not the same predicate if $C_1 \neq C_2$.

10.3 Why Non-Contextual Assignments Fail

A non-contextual hidden variable assignment demands: for every observable O, a definite value v(O) exists prior to and independent of measurement context. Under LRT this is inadmissible. A predicate's resolution is defined relative to an interaction scope. There is no context-free resolution. Demanding v(O) independent of context demands a predicate have a determinate truth value independent of the respect under which it is evaluated. This violates L_3's Identity requirement that identity be determinate relative to a specified respect.

10.4 The Derivation

UNS: for a complete state description, exactly one successor is admissible. The successor depends on the interaction scope.

JRC: for entangled systems, predicate resolution of the global state cannot be decomposed into independent local resolutions.

Together: the outcome of measuring observable O on system A depends on the global state s_AB and the interaction scope (which observables on B are co-measured). This is quantum contextuality.

$$L_3 \rightarrow \text{context-indexed predicate resolution} \rightarrow \text{no context-free value assignment} \rightarrow \text{Kochen-Specker}$$

ESTABLISHED	Kochen-Specker contextuality follows from L_3's requirement that predicates be evaluated under a definite respect (interaction context). Non-contextual value assignments violate Identity. Kochen-Specker is downstream of L_3, alongside Bell.

11. Full Derivation Chain

From		To
$A\Omega = L_3(I\infty)$	⇒	Single foundational equation — one primitive, not three
L_3 on complete state description	⇒	UNS: unique admissible successor [Theorem, derived]

From		To
UNS	⇒	Time = order over actualization [Theorem]
UNS + incomplete description	⇒	Measurement = ordinary T application [Theorem]
L_3 (Determinate Identity)	⇒	Local Tomography [Theorem 1, existing]
Local Tomography	⇒	H_AB = H_A ⊗ H_B [Theorem 6.2, existing]
H_A ⊗ H_B + Bit Equivalence	⇒	dim(H_A) ≥ 2, dim(H_B) ≥ 2 [MM4, existing]
dim ≥ 2 × 2	⇒	H_AB contains Schmidt rank > 1 states [linear algebra]
Schmidt rank > 1	⇔	JRC satisfied [JRC-Schmidt Lemma]
JRC in $\mathscr{A}$	⇒	No L_3-complete local decomposition [Non-Decomposability Theorem]
Non-Decomposability	⇒	Bell locality assumption L_3-inadmissible for entangled s_AB
L_3-inadmissibility	⇒	Bell's theorem is downstream consequence of L_3

12. Open Formalizations for Lean 4

12.1 Open 1: Formal Definition of Resolvable

The Non-Decomposability proof uses the term 'resolvable' — a predicate P is resolvable in state s. This requires a precise L_3-semantic definition that maps to Hilbert space expectation values.

Candidate definition:

> *Predicate P is resolvable in state description s iff the expectation value of the corresponding Hermitian operator M_P is uniquely determined by s alone: ⟨s|M_P|s⟩ is derivable from the description of s without reference to any other system.*

What needs to be shown: this definition is equivalent to the standard quantum-mechanical notion of a measurement outcome being determined by a local state, and that it maps cleanly to the factorizability condition in the proof.

OPEN	Candidate definition given. Requires Lean 4 formalization connecting L_3-semantic resolvability to Hilbert space expectation value determination. Until formalized, the Non-Decomposability proof contains an informal step.

12.2 Open 2: f Cannot Generate Joint Information from Locally Insufficient Inputs

The Non-Decomposability proof requires that no function f acting on two individually insufficient descriptions can produce a sufficient joint description — unless f has access to information not present in either input.

Candidate route via Identity: if f generates information about C_AB that is not present in s_A or s_B, then f is producing predicates whose grounding is not in the identity of either subsystem. Identity requires that a configuration's properties be grounded in what it determinately is, not in how it is combined. Therefore f cannot be the source of new predicate resolutions — it can only transmit what is already determined in its inputs.

What needs to be shown: this informal argument can be made a formal derivation from Determinate Identity (Id-1: intrinsic identity) without importing information-theoretic intuitions that are not themselves derived from L_3.

OPEN	Candidate route given via Id-1. Requires Lean 4 formalization. The informal argument is sound but relies on intuitions about information grounding that need to be derived from Determinate Identity rather than assumed.

12.3 Open 3: Lean 4 Proof of UNS via Leibniz Identity

The tightened UNS proof invokes Leibniz Identity of Indiscernibles applied via L_3's Identity law. Formalization requires: (1) a formal definition of the constraint set C(s_n); (2) a formal statement of Leibniz Identity within the L_3 semantic framework; (3) a proof that existence of at least one successor follows from the non-emptiness of the admissible domain.

OPEN	Tractable from existing LRT semantic framework. Not yet developed in Lean 4.

13. Epistemic Status Summary

Claim	Status	Notes
$A\Omega = L_3(I\infty)$	Established	Single foundational equation; self-certifying; one-equation theory
UNS as theorem of L_3 (tightened via Leibniz)	Established	Identity of Indiscernibles closes uniqueness gap; see Section 9
$\mathscr{T} < 1$ as epistemic underdetermination	Established	Not ontological branching
Born rule as measure over incomplete descriptions	Established	Gleason reinterpreted

Claim	Status	Notes
JRC defines L_3-irreducibility	Established	Predicate not locally resolvable
JRC $\Leftrightarrow$ Schmidt rank > 1	Established	Forward and backward proved
Non-Decomposability Theorem	Established	Sketch; conditional on Open 1-2
Bell as consequence of L_3	Established	Conditional on Non-Decomposability; see Section 7
Contextuality (Kochen-Specker) as consequence of L_3	Established	Context-indexed predicate resolution; see Section 10
LRT distinct from hidden variable programs	Established	Directional grounding; see Section 8
"Resolvable" formal L_3-semantic definition	Open	Candidate definition given; needs Lean 4
f cannot generate joint info from local insufficient inputs	Open	Candidate route via Identity; needs Lean 4
L_3 holds at Planck-scale regimes	Open	Circularity audit required throughout

Human-Curated, AI-Enabled (HCAE) — JD Longmire, author. All foundational commitments and truth-claim verification are the author's. AI used for systematic exploration, consistency checking, and exposition.